

FIRST PRINCIPLES

Access	Cheap, frequent launch — not one heroic flight. Bowen + Southern Launch exist; no Starship MoU yet. Christmas Island already recovered Ship 40 (Jul–Aug 2026).
Stay	Fleets need in-space mobility. Neumann Drive is commercial on SpaceX Transporters; debris→fuel is roadmap, not done.
Build	Print metal structures onshore (Titomic / Lab22). Traction is real aerospace/defence — not a Starship hull line.
Crew	Academy before Hollywood. Western Sydney + ADFA model; holodeck = sim first.

SYSTEMS STACK

07 Academy

W. Sydney + ADFA · sim → flight corridor

06 Medicine

AU medtech / sensing adjacency — not kits

05 Sensors

Fleet constellation · EOS SSA (Atlas = early)

04 Propulsion

Neumann commercial EP · Gilmour access · Hypersonix flown

03 Manufacture

Titomic cold-spray · Lab22 AM — pilots

02 Launch / return

Bowen orbital · Koonibba re-entry · Whalers Way

01 Foundation

TSA 23 Jul 2024 · ITAR · Statement on Space Jul 2026

WHY AUSTRALIA

- TSA in force — US vehicles can launch from AU soil (still needs licences).
- Ship 40 recovery already used AU logistics.
- Whalers Way: polar / SSO. Bowen: mid-inclination.
- Allied second-hemisphere option — not a Starbase clone (no methalox industrial base yet).

BUILD PHASES

- P1 · 12 mo — Structure demo + range diligence
- P2 · 24 mo — Academy charter (~100 cadets)
- P3 · 5 yr — Print → launch → assemble demo

HARD NO'S

- No Starship MoU. No warp.
- ELA / Arnhem not live.
- SptRIT is not Gilmour.
- Eris not orbital yet (TF2 ~early 2027*).
- Concept art ≠ hardware.

PARTNERS (FACT-CHECKED)

Gilmour (Bowen · Series E A\$217m · ElaraSat rideshare) · Southern Launch (Koonibba re-entry · A\$25m) · Neumann (Transporter cadence) · Titomic + Lab22 · Fleet · EOS · Hypersonix · UNSW / ADFA

SECTOR SNAPSHOT

Scale

A\$4.6bn turnover · ~17k people · ~620 orgs

(ASA fact sheet, Jul 2025)

Policy

Statement on Space — Jul 2026

Flight · platforms · microgravity · SSA

Capital

NRFC space-related ≈ A\$198m+

Gilmour A\$75m · Fleet Series D A\$150m

WHO'S LIVE (SEP 2026)

Gilmour

Licensed Bowen pad. Eris TF1 Jul 2025 failed (~T+9–17s). TF2 ~early 2027*. Series E A\$217m. SpaceX = rideshare only.

Southern Launch

Koonibba: commercial re-entry (from Feb 2025). Whalers Way orbital. A\$25m Series A (A\$10m NRF).

Neumann

ND-25 on Transporter-16/17. ND-50 on CarbSAR. Debris-fuel with CisLunar = roadmap.

Titomic

NASA SAA · US defence contracts · CRADA. Mostly US sustainment — not hull factory.

Hypersonix / Fleet / EOS DART AE Mach 5+ (Feb 2026). Fleet Series D >A\$800m. EOS Atlas HEL = early market, not sales.

BLOCKERS

- ITAR still applies — TSA ≠ tech transfer free pass.
- Every launch needs AU permits.
- No Starship-grade methalox chain yet.
- Thin specialist workforce.
- North range sites = diligence only.

POLICY FIT

- Spaceflight → Bowen + Southern Launch + TSA
- Platforms → Titomic / Lab22
- Trusted services → Fleet + EOS SSA
- Microgravity → Koonibba re-entry
- Defence space signal A\$9–12bn / decade — not a bid

BEFORE THE ASK HARDENS

TSA pathway · northern site shortlist · Gilmour TF2 in writing · Southern Launch readiness · Titomic P1 SOW · Academy MoU skeleton · red-team: what if Eris slips another year?

STARFLEET FOR REAL — AUSTRALIA

Kangaroo-class NCC-992-AU · SpaceX / SpaceXAI

CHIEF ENGINEER

Crystal Elle Arena-Turner

TerAustralis · Sydney · 6 Sep 2026 · 3 / 3

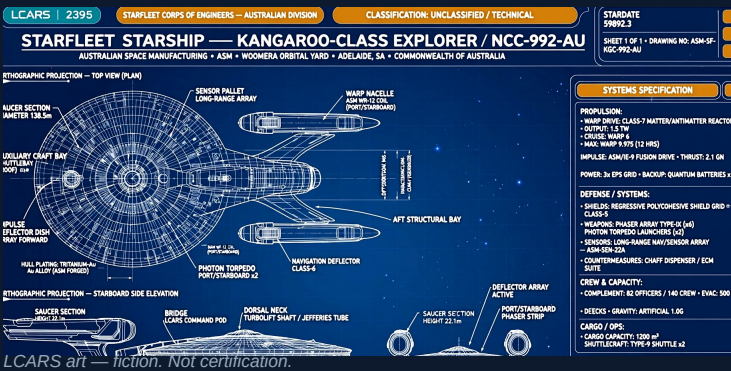
GEOGRAPHY



ACADEMY

- Y1 — Shipbuilding · propulsion · astrogation
- Y2 — UE5 / MR holodeck sims
- Y3 — Cislunar tug capstone + live range
- Spine — ADFA leadership model
- Sites — W. Sydney + Qld / SA corridors

KANGAROO-CLASS — NARRATIVE MAP (FICTION COVER)



→ Hull → Titomic / Lab22

→ Impulse → Neumann (+ Gilmour)

→ Sensors → Fleet + EOS

→ Shipyard → Bowen + Southern Launch

→ Academy → W. Sydney + ADFA

→ Recovery → Koonibba / Christmas Is.

SICKBAY ADJACENCY

Cochlear / ResMed lineage · bionic-eye research · quantum sensing. Human-systems depth — not flying devices.

MADE IN AUSTRALIA

Materials → print → launch → recover. Qld + SA corridors. North = diligence only. No dead spaceports claimed live.

STILL NEED

SpaceX BD channel · site shortlist · LOIs for P1 · Academy MoU · diligence binder (TSA, NRF, licences).

ASK

30-min exploratory — SpaceX launch / manufacturing / recovery BD (+ SpaceXAI if Grok co-travels).

access · stay · build · crew

Crystal Elle Arena-Turner · Chief Engineer · teraustralis.incognita@gmail.com · +61 450 144 997 · TerAustralis.com.au